

Strategy Overview, Data & Research Objective

1. Objective

- Build a systematic trading strategy using OHLCV data.
- Generate Long / Short / Flat signals.
- Evaluate profitability and risk under realistic execution assumptions.

2. Dataset

- Five stocks:
 - HDFCBANK
 - ICICIBANK
 - SBIN
 - AXISBANK
 - KOTAKBANK
- OHLCV variables used:
 - Open, High, Low, Close, Volume
- Training period and test period explicitly.
- Mention why chronological splitting is used: > Financial data is time-dependent, so random train-test splitting can introduce look-ahead bias.

3. Overall Framework

Data → Feature Engineering → Signal Generation → Position Sizing → Risk Management → Backtesting → Performance Evaluation

4. Core Research Hypothesis

Individual technical indicators are noisy when used independently. Therefore, the strategy combines momentum, trend, reversal, volatility and volume information to identify situations with stronger directional evidence rather than relying on a single indicator.

Strategy Development & Approaches Tested

5. Feature Engineering

Feature	Purpose	Why Used
Momentum	Price acceleration	Capture directional strength
Moving Averages	Trend confirmation	Avoid trading against

Feature	Purpose	Why Used
RSI	Overbought/oversold	broader trend Identify potential reversals
ATR	Volatility	Make risk management stock-specific
Volume Ratio/Z-score	Participation	Confirm strength behind a move
Breakout Signal	Range escape	Capture strong directional moves

6. Signal Generation

Initial approach

Single-condition technical signals.

Problem

Individual indicators produced noisy signals.

Final approach

Multi-feature scoring:

Long Score vs Short Score

- High Long Score → bullish setup
- High Short Score → bearish setup
- Neither sufficiently dominant → Flat

7. Approaches That Were Tested and Rejected

Regime Filter

Hypothesis: Strategy performance may differ between trending and sideways markets.

Result: Hard regime filtering was too restrictive / did not provide sufficient improvement.

Decision: Removed the hard regime gate while retaining volatility information through ATR.

Cooldown

Hypothesis: Re-entry immediately after a stop may create clusters of losses.

Result: Increasing cooldown reduced/rearranged trades but did not solve the underlying loss problem.

Decision: Treat cooldown as a risk-control mechanism, not as a signal-quality solution.

Short Selling

Hypothesis: Bearish signals could generate additional returns.

Result: Short trades were materially weaker in the tested universe.

Decision: Evaluate a long-only specification and compare its out-of-sample behaviour.

Backtesting Methodology & Risk Management

8. Backtesting Framework

The pipeline:

1. Calculate features using historical information.
2. Generate signals.
3. Execute on next-bar Open.
4. Determine position size.
5. Track mark-to-market P&L.
6. Apply stop/target rules.
7. Deduct transaction costs.
8. Calculate portfolio equity and performance metrics.

9. Train-Test / Warm-Up Methodology

Rolling indicators such as moving averages and ATR require historical observations before the test period begins. Therefore, features are calculated using the available historical data across the train-test boundary, after which only the designated test observations are passed to the backtesting engine.

10. Risk Management

Component	Final Approach	Reason
Stop Loss	ATR-based	Adapts to stock volatility
Target	ATR-based	Maintains defined reward/risk structure
Cooldown	Implemented	Prevent clustered re-entry
Position Sizing	Exposure constrained	Prevent excessive concentration
Transaction Cost	0.10%	Realistic simulation
Drawdown Protection	Portfolio-level	Capital preservation

11. Short Stock-wise Interpretation

HDFC: Positive OOS performance and relatively stronger consistency.

ICICI: Positive return with acceptable risk-adjusted behaviour.

SBI: Positive but relatively modest OOS improvement.

Axis: Negative result indicates weakness in signal generalization.

Kotak: Largest weakness; sharp deterioration from training to OOS performance indicates potential stock/regime sensitivity.

Combined Portfolio, Benchmark Comparison & Conclusion

12. Combined Portfolio Equity Curve

Large graph:

Date → Combined Portfolio Equity

It should represent trading all five stocks together.

Below it, optionally show:

- Initial Capital
- Final Equity
- Total Return
- Maximum Drawdown
- Sharpe
- Sortino

13. Research Findings

What worked

- Multi-feature confirmation was preferable to single-indicator signals.
- ATR-based risk management adapted better to stock-specific volatility.
- Long-only performed better than the short-enabled specification in the tested sample.
- The strategy retained positive performance on several stocks during OOS evaluation.

What did not work

- Short signals were weak.
- Increasing cooldown did not solve the fundamental signal problem.
- Hard regime filtering was not sufficiently useful.
- Kotakbank showed significant deterioration out-of-sample.

14. Limitations & Future Improvements

- Limited number of stocks.
- Limited historical sample.
- Technical features may be regime-dependent.
- Parameter sensitivity.
- Potential overfitting risk.
- Need for broader OOS / walk-forward validation.
- Need for testing across sectors and market regimes.

15. Conclusion

The project demonstrates a complete systematic trading pipeline from feature engineering and signal generation to execution, risk management and out-of-sample evaluation. While the strategy produced positive results for several stocks, the variation across stocks highlights that profitability alone is insufficient evidence of robustness. The out-of-sample analysis therefore provides both evidence of potential signal quality and clear directions for further research.